Name: _____ **Period:** _____ **Date:** _____

Topic 2.1 Guided Notes: Perception

Unit 2: Cognition | AP Psychology

Directions: Read through the topic overview page below and fill in each blank as you go. Use the tables and activities to check your understanding, then head to the quiz link at the end to test yourself.

Source page: appsychreviewhub.com/overviews/topic2.1

Part 1: Making Sense of the World (Two Processing Speeds)

Word Bank

Bottom-Up Processing	*Top-Down Processing*	*Schemas*
Perception	*Context*	*Perceptual Set*

Sensing the world is only half the battle — our brains must actively filter, organize, and interpret raw sensory data to create a meaningful reality, a process known as _____. Because this relies heavily on our memories, expectations, and biases, the reality we experience is a mental construction, not a perfect recording of the world.

- _____: starting with raw, incoming sensory data and building up to a complete perception — like blindly feeling your way around an unfamiliar dark room.
- _____: relying on your prior knowledge, expectations, and _____ (mental frameworks organized by past experiences) to interpret what you're experiencing — like navigating your own dark bedroom effortlessly.
- _____: a mental predisposition to perceive something a certain way based on recent experiences or expectations — like interpreting every creak as sinister after a friend warns you a hallway is haunted.
- _____: the surrounding information that allows your brain to guess what something (like a friend's messy handwriting) is supposed to mean before your eyes even fully process it.

Your cultural background shapes these expectations too — someone raised in a culture where direct eye contact signals respect might perceive the exact same behavior very differently than someone raised in a culture where averted eyes are the sign of respect. The raw sensory information is identical; the culturally shaped perception is not.

Apply It: Which Type of Processing?

Using the word bank above, identify which concept best matches each scenario.

#	Scenario	Concept
1	A woman walks confidently through her own kitchen in the dark, immediately knowing where the counter and refrigerator are without needing to feel around.	
2	A pharmacist can read a doctor's messy handwriting because the rest of the prescription's wording lets her guess the scribbled word before she even fully looks at each letter.	
3	A child raised believing spiders are dangerous flinches at the sight of a small, harmless spider, while a child raised calmly handling spiders picks it up without fear.	

Part 2: The Filters (Attention and Inattention)

Word Bank

Change Blindness	*Attention*
Inattentional Blindness	*Selective Attention*

Millions of sensory inputs hit your receptors every second. Your brain uses _____ to focus awareness on specific stimuli while ignoring the rest.

- _____ (the "Cocktail Party Effect"): tuning out competing stimuli, like the music and chatter at a loud party, to focus entirely on the person in front of you.
- _____: completely missing information sitting in plain sight because your brain is pouring all its cognitive resources into one thing — famously demonstrated by the "invisible gorilla" study, where participants counting basketball passes fail to notice a gorilla walk through the scene.
- _____: failing to notice a significant difference between two versions of a scene, rather than missing something within one continuous view — like not noticing when a researcher is swapped for a different person mid-conversation.

Apply It: Which Attention Concept?

#	Scenario	Concept
1	A magician's sleight-of-hand trick works because the audience is so focused on his waving right hand that they completely miss his left hand secretly palming a card.	
2	During a video call, the person you're talking to is secretly replaced by a different person wearing similar clothing mid-conversation, and you don't notice the switch at all.	
3	A student studying in a busy coffee shop tunes out nearby conversations and background music to focus entirely on the audiobook playing in her headphones.	

Part 3: Gestalt Principles (The Whole Is Greater Than the Parts)

Word Bank

Figure-Ground	*Gestalt Principles*	*Proximity*
Closure	*Similarity*	

Early German psychologists noticed that people have a natural tendency to organize visual elements into meaningful wholes — you perceive a unified "car," not a random pile of glass, rubber, and metal. They called these grouping tendencies the _____.

- _____: your brain naturally organizes the visual field into objects that stand out from their background — like focusing on a teacher speaking at the front of a busy classroom.
- _____: your brain groups elements based on shared visual features, like automatically grouping soccer players by jersey color.
- _____: your brain groups elements that are physically close together, like perceiving shoulder-to-shoulder defenders as one unified wall.
- _____: your brain mentally fills in gaps to perceive a complete whole, like reading a neon sign as a full word even when a few bulbs are burned out.

The diagram linked on the study guide page shows a fifth principle alongside these four — continuity — though it isn't spelled out in the text above, so check the video and diagram for it.

Apply It: Which Gestalt Principle?

#	Scenario	Principle
1	Despite scattered clouds of many different shapes in the sky, a person quickly picks out all the fluffy white ones as belonging to one group, separate from the thin gray ones.	
2	A viewer instantly focuses on a single red balloon rising against a plain blue sky, perceiving the sky simply as empty background.	
3	Even though a neon sign is missing several lit-up letters, a driver still easily reads the complete word it's supposed to spell.	

📺 Video Notes: Perceiving is Believing (Crash Course Psychology #7)

Answer the questions below as you watch — some ideas will already be familiar from Parts 1 and 3, and that's fine; reinforcing them is the point.

- How does the video use the duck–bunny image to show that expectations shape perception, and what does it mean by “believing is also seeing”?

- What examples does the video give of context, culture, and emotions or motivation shaping what we perceive?

- The video names three rules of grouping: proximity, continuity, and closure. What does each mean, and what example does the video give?

- How does the video explain retinal disparity and why it doesn't help much with far-off distances? Which four monocular cues does it describe, and what example goes with each?

- What does the upside-down face demonstration show about where perception happens, and how does the video's Lego analogy describe the relationship between the senses and the brain?

Part 4: Perceiving Depth and Motion

Word Bank

Texture Gradient	*Autokinetic Effect*	*Phi Phenomenon*
Retinal Disparity	*Monocular Depth Cues*	*Convergence*
Linear Perspective	*Relative Size*	*Binocular Depth Cues*
Interposition	*Stroboscopic Motion*	*Perceptual Constancy*
Relative Clarity		

Your retinas are flat, only receiving a 2D image of the world. Your brain calculates 3D depth using two different toolkits.

The first requires both eyes: _____. Your brain compares the two slightly different images each eye receives — a cue called _____ (exactly how 3D movie glasses trick your brain). Your brain also uses _____, a muscular cue where your eyes physically turn inward as an object gets closer, using that muscle strain to calculate distance.

The second toolkit only requires one eye: _____. If two objects are similar in size, the one casting a smaller retinal image is perceived as farther away (_____). Hazy or blurry objects look distant compared to sharp ones (_____), and coarse, detailed textures appear closer than smooth ones (_____). If one object partially blocks our view of another, the blocking object is perceived as closer (_____). Finally, parallel lines (like train tracks) appearing to converge in the distance signal depth (_____).

Once depth is calculated, your brain uses it to maintain _____ — perceiving an object as stable in size, shape, color, or brightness even as the raw image on your retina keeps changing (like a swinging door still looking rectangular despite casting a shifting, trapezoidal image).

- _____: rapid flashes of slightly different still images create the illusion of smooth motion, like in an animated movie.
- _____: two or more separate lights blinking on and off in quick succession create the illusion of a single light smoothly moving between them, like a theater marquee.
- _____: a single stationary point of light in a pitch-black room will eventually appear to drift or float, simply because your brain has no other visual reference points to lock it in place.

Apply It: Which Depth Cue or Motion Illusion?

#	Scenario	Concept
1	A viewer wearing special 3D glasses at the movie theater perceives objects flying off the screen because each eye is being fed a slightly different image to compare.	
2	Standing at the start of a long, straight set of train tracks, a photographer notices the two parallel rails appear to meet at a single point far in the distance.	
3	A parked car partially hidden behind a large truck is perceived as being farther away than the truck.	
4	A cartoon character appears to smoothly run across the screen, even though the animator actually only drew a rapid sequence of separate, slightly different still images.	

5. Don't Trip Up! (Common Misconceptions)

Think of assembling a jigsaw puzzle. Bottom-Up is trying to put the puzzle together without ever looking at the box cover (building piece by piece). Top-Down is looking at the box cover first, so you already have an expectation of what the final picture should look like before you place the pieces!

_____ is about the visual image difference hitting the retina. _____ is about the physical muscle feeling of your eyes turning inward. Both require two eyes, but they rely on totally different mechanisms.

6. Quick Check: Review Questions

Circle the best answer for each question below.

1. Standing in the pitch-black hallway of a house she has never visited before, a woman must feel the texture of the wall and the edge of a doorframe, slowly piecing these raw sensory clues together to figure out the layout of the space. This is an example of:

A. Bottom-Up Processing
B. Change Blindness
C. Perceptual Set
D. Top-Down Processing

2. Right before walking down a hallway, a woman is told by a friend that the space is haunted. As she walks, she interprets every small creak in the floorboards and every shifting shadow as evidence of something sinister. This mental predisposition to interpret a stimulus in a certain way is called:

- **A.** Sensory Interaction
- **B.** Schema
- **C.** Selective Attention
- **D.** Perceptual Set

3. At a loud, crowded party, a woman is able to completely tune out the music and dozens of overlapping conversations to focus entirely on what her friend directly in front of her is saying. This ability is known as:

- **A.** Change Blindness
- **B.** Selective Attention (The Cocktail Party Effect)
- **C.** Inattentional Blindness
- **D.** Perceptual Constancy

4. In the famous "invisible gorilla" study, participants who were asked to carefully count the number of basketball passes in a video frequently failed to notice a person in a gorilla suit walk directly through the middle of the scene. This failure to notice an unexpected object because attention was focused elsewhere is called:

- **A.** Change Blindness
- **B.** The Phi Phenomenon
- **C.** Perceptual Set
- **D.** Inattentional Blindness

5. During a free kick, a spectator's brain automatically perceives the defenders as a single unified wall because they are standing shoulder-to-shoulder, tightly packed together in the same physical space. This Gestalt principle is called:

- **A.** Similarity
- **B.** Closure
- **C.** Proximity
- **D.** Continuity

6. As a baseball flies closer and closer to a catcher's face, his eyes physically turn inward to keep tracking it, and his brain uses the resulting muscle strain to help judge exactly how close the ball is. This binocular depth cue is called:

- **A.** Relative Size
- **B.** Convergence
- **C.** Retinal Disparity
- **D.** Texture Gradient

7. In a photograph of a gravel road, the individual pebbles near the bottom of the frame look sharp, distinct, and highly detailed, while the same pebbles farther up the road blur together into a smooth, indistinct surface. This monocular depth cue is called:

- **A.** Texture Gradient
- **B.** Interposition
- **C.** Relative Clarity
- **D.** Relative Size

✅ Check Your Progress

Done with your notes? Head over to the Quiz to test what you've learned:

https://appsychreviewhub.com/quiz

Source content adapted from Psych Review Hub — Topic 2.1: Perception (appsychreviewhub.com)